

Reviewer Response Status — IEEE Access 33821

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Contents

All comments in one table
Reference numbers

12 comments, one-shot resubmission. One row per comment: what was asked, what changed, and what is still open. Manuscript sources: paper-access.tex (A) and paper-build.tex (B), kept in sync. Work line: main.

All comments in one table

Comment	Asked	Done	Remaining
R1-1 single collection, 31 images	Only 31 hand-verified images from a single collection, with wide confidence intervals: no proof the method works on other drawing styles or at larger scale, and the hand-verified labels themselves carry possible residual bias.	Results labeled as within-collection only, with drafter groups presented descriptively (16 direct and 15 inferred assignments disclosed, per-drafter table included); no broad claims and no held-out-drafter experiment; residual label bias disclosed as mitigated but not eliminated, with independently authored synthetic data as the second leg; future expansion described as future work.	Owner read-through

Comment	Asked	Done	Remaining
R1-2 complex parts	Which complex devices are really handled: pin modeling for non-trivial parts, and switches currently exported as plain resistors.	New capability table separating tested connectivity, estimated pins, and illustrative export, with a metric-boundary row; switch behavior stated exactly as the code does it (midpoint pins, conditional 0.001-ohm emission).	Owner read-through
R1-3 hand-set thresholds	Fixed scale-relative thresholds set by hand, including the alpha the reviewer names; no adaptive learning; behavior at size extremes unknown.	All values stated plainly (alpha 0.35 fallback-only with formula, multipliers 0.62/0.30/0.20, clamps, 4x reach, macro sweep 0.895-0.903); untested extremes admitted.	Nothing
R1-4 no values, can't simulate	No component values are read, so the output cannot simulate despite SPICE-format claims.	Scope stated in abstract, intro, and conclusion; retitled including heads and diagram; export labeled illustrative throughout, external values required.	Nothing
R1-5 small circuits only	Narrow coverage of 3 to 14 supported parts with median 7; dense buses and multilayer crossings untested.	Limits stated; true distribution printed in the caption (15 at most 5, 12 at least 10, median 7, range 3-14); nothing claimed beyond it.	Nothing

Comment	Asked	Done	Remaining
R1-6 memristor citations	Cite two suggested papers (memristor neural circuit with classical conditioning; memristive decision-making for navigation) in the SPICE-simulation sections.	Both checked and cited in Related Work with full references (IEEE Trans. Consumer Electronics, Feb 2026; IEEE Trans. Industrial Informatics, vol. 22).	Template check
R2-1 statistics, roadmap	Statistical treatment of the results plus a research roadmap; historical timing data missing.	Sampling in four strata with independent annotation and adjudication described; roadmap added; missing timing honestly noted.	Owner read-through
R2-2 VLM comparison	Whether the VLM comparison is fair (it receives annotated boxes) and how cost is framed.	Labeled as a diagnostic with identical annotated inputs (21/31 exact on both sides); no equivalence or cost claims.	Nothing
R2-3 title overclaims	Title promises simulation-ready SPICE output the method does not produce.	Retitled everywhere, including running heads and the pipeline diagram; metric boundary stated.	PDF rebuild
R2-4 crossing wires	Whether crossings get merged into junctions, including detector misses, and what that costs netlist correctness.	Measured tradeoff reported in full (27/4/0 goes to 19/0/8, stored artifact kept); removing edges also removes true connections, so no gain claimed; detector-miss part honestly marked unmeasured.	Nothing possible

Comment	Asked	Done	Remaining
R2-5 ablations	Module ablations judged severely insufficient as evidence for each part's contribution.	Real table as observed, including fixed-pixel base 0.820 vs scale-relative 0.816 and full-pipeline ties at 0.890 presented as-is without spin; missing full-occlusion and scale tests disclosed.	Nothing provable
R2-6 single-scalar limit	Discussion point on the limits of single-scalar evaluation.	Kept as-is; no change needed.	Nothing

Shared language everywhere: a non-finding is not equivalence; no blanket guarantees; pair scores do not certify pin layouts; lab-condition results do not prove real-world dominance.

Reference numbers

Join 0.890 (418/37/66) on all 31; perfect-wire 0.88983 (420/40/64). Wire ledger: 0.9755 = 10°/18px config, 0.9726 = 12°/8px production. Otsu 0.789. VLM 0.923, 21/31 exact. Reach sweep macro 0.895–0.903. Numeric gate: 192/192 stored comparisons pass.